Load flow study is carried out for

1. Fault calculation
2. Stability studies
3. System planning
4. Load frequency control

Q1. The most appropriate operating speeds in rpm of generators used in Thermal, Nuclear and Hydropower plants would respectively be

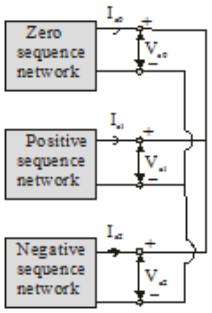
A. 3000, 300 and 1500

B. 3000, 3000 and 300

C. 1500, 1500and3000

D. 1000,900and750

The connection diagram of sequence networks for a particular fault on a power system network is given in the figure. The type of the fault is



A. Single line to ground

B. Double lines to ground fault

C. Line to line fault

D. Open circuit

41. Lightning arresters are used in power system to protect electrical equipments against

A. Direct strokes of lightning
B. over voltages due to indirect lightning stroke
C. Power frequency over-voltages
D. Over currents due to lightning

The incremental fuel cost for two generating units given by

$IC_1 = 25 + 0.2\ PG_1$

$IC_2 = 32 + 0.2\ PG_2$

Where PG_1 and PG_2 are real power generated by the units.

The economic allocation for a total load of 250 MW, neglecting transmission loss is given by:

- A. PG₁ = 142.5 MW; PG₂ = 107.5 MW
- B. PG₁ = 109.75 MW; PG₂ = 140.75 MW
- C. PG₁ = PG₂ = 125 MW
- D. PG₁ = 100 MW; PG₂ = 150 MW

Right Answer is: A

SOLUTION

For economical load, $IC_1 = IC_2$

$25 + 0.2\;P{G_1} = 32 + 0.2\;P{G_2}$

$\Rightarrow P{G_1} - P{G_2} = 35$

Given that total load is 250 MW

$\Rightarrow P{G_1} + P{G_2} = 250$

$P{G_1} = 142.5\;MW,\;P{G_2} = 107.5\;MW$

Transmission lines are transposed to

A. reduce copper loss

B. reduce skin effect

C. prevent interference with neighbouring telephone lines

D. prevent short-circuit between any two lines

In the solution of load-flow equation, Newton-Raphson (NR) method is superior to the Gauss-Seidal (GS) method, because

- A. the time taken to perform one iteration in the NR method is less when compared to the time taken in the GS method.
- B. number of iteration required is not independent of the size of the system in the NR method.
- C. Number of iteration required in the NR method is more when compared to that in the GS method
- D. Converges characteristics of the NR method are not affected by the selection of slack bus.

1. Zero sequence currents can flow from a line into a transformer bank if the windings are in

A. ground star/ delta

B. delta/ star

C. star/ grounded star

D. delta/ delta

Correct Option: A

4. In the optimum generator scheduling of different power plants, the minimum fuel cost is obtained when

(a) only the incremental fuel cost of each plant is the same

(b) the penalty factor of each plant is the same

(c) the ratio of the incremental fuel cost to the penalty factor of each plant is the same

(d) the incremental fuel cost of each plant multiplied by its penalty factor is the same.